

$a + ? m \# 2 [\pi \text{ } \text{ } 3 :$

“o ,ÆÊ)•I' úùØ³ a+?m# 4Ti,, @™¥ 'ÀQb

9XjX Zm MimK *Q?QKQHq;v Q7 :`TbB K MMBbMbfXQpbi2/ "m+?
`sBp?iiTb,ff `tBpXQ`;f #bfK i?fyRyeke3

Mz , kyyR

É– , 3:

,• D ,

ó :` bbK MMB]Nöi¥e†£ü

£ü 0 SB2`B, 0 :B K#2HTHB

ó 6mHiQM@qQ Q¥r£ü

K1 , 'Óó :` bbK MMB ]Nöiö1çØ¥e±£ü “ "2`i` K

0 SB2,`BB K#2HTHBü

1oM , [m MimK +Q?QKQHq;v- :` bbK MMB Mb- SB2`B 7Q`KmH -

9X9X *2`i BM A/2MiBiB2b QM i?2 Zm MimK S`Q/m+i Q7 a+?m#2

R3 RX :_ > J SPaAhAoAhu P6 h_AS G1 a*>I"1_h * G*IGIa

$\hat{I}[S_2 \ddot{I}$

$$S_{12}(t) = 1; \quad S_{21}(t) = x_1$$

è RX8 Ua+?hT#2`S₃ ¥>œ9Ø WX S₃ Ĩµ 3!=6 ñÂĐbáìP""µØ
0¾w,, ai MH℘v[TZ7Ÿ9ØÄñ a+?m#ȚTb

$$SHqĐ"µØ0 \ X \acute{K}É\hat{A}Đ w_0=321 \textcircled{R} SHq$$

$$UdV \quad S_{321}(t) = x_1^2 x_2:$$

$$"\mu\varnothing0 @ \zeta l^1 @ (f) = \frac{f}{x_i} \frac{s_i(f)}{x_{i+1}} \ddot{I} \quad s_i \mathfrak{E} \mathfrak{D} x_i \text{,,} x_{i+1} b$$

$$\acute{c} \uparrow [T @ \not\equiv T'' ? 5^1$$

$$U3V \quad @ (x_i) = 1; \quad @ (x_{i+1}) = 1;$$

$$UNV \quad @ (x_i^k) = x_i^{k-1} + x_i^k {}^2x_{i+1} + \quad + x_{i+1}^{k-1}; \quad @ (x_j) = 0 \quad (j \notin i; i+1);$$

$$URyV \quad @ (1) = 0:$$

$$\acute{c} \acute{E} \text{¾}w9\varnothing \ Xai2T \ R(w) = 0 \ \acute{s} \textcircled{C} \hat{A} \mathfrak{D}$$

$$URRV \quad w = 123: \quad S_{123}(t) = 1:$$

$$ai2T \ k(w) = 1$$

$$\acute{c} \quad \grave{ }(w) = 1 \ \not\equiv \hat{A} \mathfrak{D} \quad a+?m\#\mathfrak{T}i\textcircled{R} \quad G2?K\mathfrak{Z}\grave{ }$$

$$URkV \quad w = 213 \quad (G2?K2(1;0;0)): \quad S_{213}(t) = x_1;$$

$$URjV \quad w = 132 \quad (G2?K2(0;1;0)): \quad S_{132}(t) = x_2:$$

$$ai2T \ j\grave{ }(w) = 2$$

$$\acute{c} \quad \grave{ }(w) = 2 \ \not\equiv \hat{A} \mathfrak{D} \quad P""\mu\varnothing0 \ \mathfrak{E} b$$

$$w = 231 \quad : \quad \grave{ } \quad b b K \quad M M \mathfrak{B}(w\mathfrak{Y}) = f1g$$

$$UR9V \quad S_{231}(t) = @ (S_{321}(t)) = @ (x_1^2 x_2);$$

$$UR8V \quad @ (x_1^2 x_2) = \frac{x_1^2 x_2}{x_1} \frac{x_2^2 x_1}{x_2} = x_1 x_2;$$

$$UReV \quad) \ S_{231}(t) = x_1 x_2:$$

$$w = 312 \quad d \quad : \quad \grave{ } \quad b b K \quad M M \mathfrak{B}(w\mathfrak{Y}) = f1g \quad P'' \quad ai \ MH\mathfrak{X}v[TZ7 \quad S_{312}(t) =$$

$$x_1 + x_2 b$$

$$\mathfrak{E} \quad S_{312} \wedge \grave{e} \ [Tb$$

$$ai2T \ \mathfrak{g}(w) = 3 \ KÉ\hat{A} \mathfrak{D}$$

$$URdV \quad w = 321: \quad S_{321}(t) = x_1^2 x_2 \quad SHq \quad :$$

S₃ a + ? m # 2 [π ø 9 V X Û V • 9 Ø S₃ ¥ a + ? m # 2 T î Â /
w

R X ! © M

k j

$$\begin{aligned} \ddot{I} \quad S^{2m+1} \wedge 2m+1 \gg o \ddot{e} \quad X \\ \div 8^1 \quad - \quad \grave{c} \quad T = (C)^n - \mu, \\ ET = (S^1)^n = \underset{m!1}{H} B^2 K \end{aligned}$$

çl R X k q2vH ;`Q m 2 çl Ñ Ü ë H = f H 2
 h: (H)=0g b 1 ç f ñ Ñ Ü ë ¥ Q s T " h b
 ® î µ Q s 2 3 î ¥ µ K • ë¹ q2vH q2vH ;`Q m: T W b
 gl_n W = S_n i i Â Ð • Ä ñ Â Ð û ^ t ' Q ¥ ð b ° 4 S_n í •
 U S à - W ¥ ë ÿ b

gl_n ï e † Q s_i ° Æ Ð M # ¥ i „ i + 1 U S à b f Ð * Q t 2 i 2 ¥² > † B
 Á b

l^{1/2} " q2vH í ¥ Ê Â l '
 ó ç B ñ q2vH í w 2 W = S_n ñ Â ... ó T " ó ô " A \$
 çl R X k l 1/2 U Ð d l 1/2 " V X ç w 2 S_n ç l
 l^{1/2} " l(w) = f(i; j) : 1 i < j n; w(i) > w(j) g
 d l 1/2 " J(w) = f(i; j) : 1 i < j n; w(i) < w(j) g = l(w₀ w)
 ï w₀ = (n; n 1; ::; 1) ^ K É Â Ð b
 + ... l(w) : c w Ü ' t " 1/2 ó J ó b j l(w) j = `(w) ž z ^ w ¥ É
 " ` m ? 1/2 ï ¥ Ö b

0 • N (w) V @ TM ž @ TM ¥ i ^
 K ^a á ì ° @ TM + ... ï ¥ 1 o 0 • b

çl R X k q • U N (w) Ð B (w) V X Ä ñ w 2 W ç l

$$N(w) := N \setminus w N w^{-1}$$

"

B (w) := T N (w)
 + ... N (w) ^ N ¥ B ñ 0 [» " ¹ j l(w) j b
 ô > m K T ? ` 2 w l a 2 + i B Q M <sup>1</sup> [ 3 ) N (w) ] ç a + ? m # B ` i B w B / B =
 C ^(w₀) `(w) Y V x 7 ! x w B / B b + Y ¹ N (w) ^ ð Y ¥ b

kX,y

' á ì ù î @ TM F l_n (C) ¥] Ø ì H B ñ ' Ù 5 ^ ñ a + ? m # B Mi ð ö
 ¢ ž l ¹ \$ ÷ Ú ' ^{1 a} Â T á ì " x a + ? m # 2 Ti S_w (t; v) T ¹ @ TM © M]
 Ø Ê ¥ } V í N_{* 1} ð S_u (t; v) S_v (t; i) Â ... Z 7 ¹ † ñ S_w (t; i) ¥ L Ÿ F † \$ Z 7
 " " c_{u,v}^w (v i) i @ l ¹ Ÿ É \$
 f „ • á ì ù î [/ Z 7 T

$$S_u(t; v) S_v(t; i) = \sum_{w \in S_1} c_{u,v}^w(v i) S_w(t; i)$$

ï c_{u,v}^w (v i) ^ 1 ç v „ i ¥ [T b R y

N x a + ? m # 2 Ti ^ G b + Q m t @ a + ? C E 2 M # 2 < ; 2 E - " ¥ v M Ÿ : c @ TM ¥
 + y Ä • b
 R y v = i = 0 H c_{u,v}^w (0; 0) 2 N ^ Ü Ä ¥ a + ? m # 2 È " 9 " @ TM ï a + ? m # B ¥ Æ Ä ñ
 " i i ' > B H # 2 È ð Ù 5 n b 2 + i B Q M • b

k3 RX :_ > J SPaAhAoAhu P6 h_AS G1 a*>I"1_h * G*IGIa

kXRy+yzBî Øñ% X¥ Ä XùîfñÛ5¥Bñ1-!{^VKet¥f
f7S -^ai„w<b
»B„ :`bbK MMB™Mb' u„ vû^ :`bbK MMBHMYfM^a+Y
b b k @ :`bbK MfMBwM¥ a+?m# 2Tiv[•îLÿy0¥ð S_w(t; y) =
Q_k^k_{i=1}(x_{a_i} y_{b_i}) fÛy0Ä™TP^að Z7¥² B" -b JQH2p@dk)nMð
ft"¥F† T EMMibQM^að Q (..³db fÛ B/ :`? K3
£ü [/žÿ²T

çØ RXR U: `? K SQbBiBpBiv h??2Q`2KXjXk;W2 S_n sY^ k₁@
:`bbK MMBk<sub>2</sub>M@:`bbK MfMBM

$$c_{u,v}^w(v_i)2N[t_i y_j]_{ij}1$$

'Z7""^1¿ t_i y_j ¥dµœ"" [Tb ^{RR}
»=„ €Û :`bbK MMBk<sup>a</sup>M Bñ1-¥Û5^ :`? K¥žÿ²T?žw
<ž©it\$ a Km2H4 [/‰Xii''Ó¥ö1²T h?2Q`2K RXR

çØ RXkUh?2Q`2K R%RX,(kaR KQMB2+inx! 2u;R;Xv2)S 5 c_{u,v}^w(v_i)2
N[t_i y_j]_{ij}1b

´^aÿi¥^ " •f™ " b2T`i2/2b'+2Mitb2(b) KBM (v)bnç
l RX9™9\$ sùîb a Km2H4yëžÿT 6M:mQ»=T€ (8)P`
TmxxH2b:`Qi?2M/B[2¥S¶=ó 6Bñ Tb
»Ø„ Ket¥+èb ÂTáì a Km2H4Kï| v= i=0 5""|Ä¹ôY
dµœ"bNHZ7TM¹

$$S_u(t; y) S_v(t; y) = \sum_w c_{u,v}^w(0; 0) S_w(t; y)$$

fž^ EB`B%oHXQp, RX'R \*Q`QHH ÿv•¥K™

w, RXR U*Q`QHH `v R%X,(REB`QBHD2+piX`2 B;)W2 S_n µ
@_{w/v}S_u(t)2N[t]b

fÚ@_{w/v}^J+/QM Rv„(E¥ bF2'rµØ0 Ð 6QKBM@EB`B%oHXQp
1b

ÂmîU Øñ% X¥HqGQhÐiiV1p u;vû^ :`bbK MMBk^aM
žKet¥ v= i=0 f™ b

‰X¥ ^a Q	Hq	}VçØ
»B„	u;vû^ :`bbK MMBM :`? K SQbBiBpBiv	
»=„	u;v©it	a Km2H4
»Ø„	v= i=0	EB`B%oHXQp

'Ó¥£üG ¿ÚÄñ'¥ :`? KžÿçØ çØ RXjë "" c_{u,v}^w(v_i) ¥
•+...³d R^kT1-ÁÖÎó "BHH2v(j)İžÿ¥+...³db
áì[1¿ :`Qi?2M/B[2¥F‰X²•'Ó¥), n b2+iBQBM

kXkçØ¥è0 X¹úù €yë°4†` áìGQ £ØñçØ¥žÿ², Ä
„û,{W%9Øb

R_U1|1 :`? K SQbBiBpBivbbK MMB\$Mk<sub>1</sub>;k<sub>2</sub> ¥T`^I¹\$n,c b2+iBkBM
R_kÐ EMMibQM^að Q³(d,jiiWn,c b2+iBkBM

k X j X : ` ? K S Q b B i B o k B i v X

è R X R c 9 O U R X R i : ` b b K M M B ¥ M W X 1 I 1 : ` b b K M M B † M †
y \$ Â - î • k @ : ` b b K M M B w M a + ? m # 2 T i S w (t ; v) V [• î L Y y 0 ¥
ð

$$S_w(t; v) = \sum_{i=1}^k (x_{a_i} \quad y_b)$$

f Õ y 0 Ä T M T P a ð Z 7 ¥ 2 + Y b i i á ì ° 3 | F L Y y 0 M ð - a x
• æ Ø ' V b

8 è 0 u = 21 v = 12 21 ^ : ` b b K M M B 1 M ^ š © Â Ð b
° V a a ï x a + ? m # 2 T i n è R X d

$$S_{21}(t; v) = x_1 \quad y_1; \quad S_{12}(t; i) = 1$$

9 Ø ð

$$S_{21}(t; v) \quad S_{12}(t; i) = (x_1 \quad y_1) \quad 1 \\ = x_1 \quad y_1$$

N H Z 7 " " c_{21;12}^{21} = 1 \, 2 \, N \, f \wedge B \tilde{n} \ddot{U} O \pounds \check{z} \ddot{Y} \pounds b
÷ B î ¥ è 0 u = 132 v = 213 ñ û ^ : ` b b K M M B M
° V a

$$S_{132}(t; v) = (x_1 \quad y_2)(x_2 \quad y_2); \quad S_{213}(t; i) = x_1 \quad t_1$$

9 Ø ð

$$S_{132}(t; v) \quad S_{213}(t; i) \\ = (x_1 \quad y_2)(x_2 \quad y_2)(x_1 \quad t_1) \\ = (x_1 x_2 \quad x_1 y_2 \quad x_2 y_2 + y_2^2)(x_1 \quad t_1) \\ = x_1^2 x_2 \quad x_1 x_2 t_1 \quad x_1^2 y_2 + x_1 y_2 t_1$$

$$U Z 7 V \quad x_1 x_2 y_2 + x_2 y_2 t_1 + x_1 y_2^2 \quad y_2^2 t_1$$

æ " t_1 = (t_1 \quad y_1) + y_1 | c \quad t_1 \pounds [\cdot s \quad V \pounds \hat{i} \mu " " \hat{u} ^ \quad t_j \quad y_i \pounds d \mu \pounds " "
" [T b R_j
^2, ' u; v \hat{u} ^ : ` b b K M M B H M ð Z 7 " " " L \hat{u} ^ (t_j \quad y_i) \pounds d \mu \pounds
" " [T f \pounds \check{\varsigma} Ø R X R y b

k X 9 X a K % X H H q h Ð € Õ : ` b b K M M B p M X

è R X R c 9 O U R X d : ` b b K M M B ¥ M W X Ø R X 2 è ' P u; v , ^ : ` b b @
K M M B † M Z 7 " " " - û ž Y b 1 a ü f B Ä á ì n B ñ d : ` b b K M M B M
¥ è 0 b

d : ` b b K M M B † M u = 231 ¥ 2 u = 231 ¥ " F 1 (2; 3; 1) µ ñ / 2 b + 2 M i b
2 > 3 „ 3 > 1 y N , ^ : ` b b K M M B V M

$$S_{231}(t; v) = x_1 x_2 \quad x_1 y_3 \quad x_2 y_3 + y_2 y_3$$

ÿ i f , ^ L Y y 0 ¥ ð 7 ^ = Q [T i i f ž ^ Ð : ` b b K M M B M É u
Y b

jy RX :_ > J SPaAhAoAhu P6 h_AS G1 a*>I"1_h * G*IGIa

| v=312 9^d : ` b b K M M B M

$$S_{312}(t; i) = x_1 \quad t_3$$

» B „ Z 7 ð ¿ † [T Z 7

$$\begin{aligned} S_{231}(t; y) \quad S_{312}(t; i) \\ = (x_1x_2 \quad x_1y_3 \quad x_2y_3 + y_2y_3)(x_1^{x_1} \end{aligned}$$

$NX8\%w\bullet\text{„}\zeta\text{ } .2\text{`Bp iBQM a},2\text{`TbR „}\mathcal{E}\sim Fb$
 $\zeta\text{I } F=(ev_1;ev_2):\overline{M}\text{! } X\text{ } X\text{ } f^{\wedge}\text{ } T\text{ } @MOa'\text{ }\mathbb{Y}\text{ } T\text{`QT}2\text{`b}$
 $\textcircled{R} (RNG2KK\text{ } e\mathbb{V}_3\text{ } Y(w))\text{ } ^{\wedge}\text{ } T\text{ } @çay\ddot{A}a,V\ddot{y}s\text{! } B\text{``2/m+B\#H2+QKTQM2}$
 $\mathbb{Y},\mathcal{E}i\text{ } \ddot{A}\ddot{n}s\text{! } \ddot{A}\gg\text{`}^1\text{ } l(w)b$
 $\mathfrak{a}\text{` } ev_1(u)^T\text{ } ev_2(v)^T=(F^T)[X(u)\text{ } X(v)]_T\text{ } V|VrT\times\bullet^1$

$$c_{u,v}^{w;d}=(\text{ } x\text{ } x)^TF^T\text{ } (F^T)[X(u)\text{ } X(v)]_T\text{ } (ev_3^T)e(w)^T\text{ }:$$

$\text{„}\zeta\text{ } k<\text{` } S\text{`QD2+iBQM } \mathbb{Q}\text{`KmH}$
 $\textcircled{R}\text{ } T\text{`QD2+iBQM } 7Q\text{`KmH}$

$$F^T\text{ } (F^T)[X(u)\text{ } X(v)]_T\text{ } (ev_3^T)e(w)^T=[X(u)\text{ } X(v)]_T\text{ } F^T\text{ } (ev_3^T)e(w)^T\text{ }:$$

$\text{„}\zeta\text{ } j\text{)}\emptyset\text{ } ev_3\text{ } \mathbb{Y}\delta^{\wedge}b$
 $\textcircled{R} (RNG2KK\text{ } e\mathbb{V}_3\text{ } Y(w))\text{ } ^{\wedge}\text{ } T\text{ } @çay\ddot{A}a,V\ddot{y}s\text{! } V_i\text{ } \mathbb{Y},\mathcal{E}i\text{ } \ddot{A}\ddot{n}s$
 $\text{! } \ddot{A}\gg\text{`}^1\text{ } l(w)b$
 $<\text{` } Tmb?\text{ } @7Q\text{`r` } \text{`/ } 7Q\text{`KmH}$

$$F^T\text{ } (ev_3^T)[Y(w)]_T=\sum_i^Xa_i[F(V_i)]_T;$$

$\ddot{I}\text{ } a_i\text{ } ^{\wedge}\sim Fj_{V_i}:V_i!\text{ } F(V_i)\text{ } \mathbb{Y}Q\text{`}b$
 yN

$$c_{u,v}^{w;d}=\sum_i^Xa_i(\text{ } x\text{ } x)^T([X(u)\text{ } X(v)]_T\text{ } [F(V_i)]_T):$$

$\text{„}\zeta\text{ } 9<\text{` } \text{` } ?\text{ } K\text{ } SQbBiBpBiv$
 $\ddot{A}\ddot{n}\text{ } [F(V_i)]_T\text{ } \textcircled{R}\text{ } \text{` } ?\text{ } Kç\emptyset\text{ } (RN^*Q\text{`QH}\text{H}\text{` } v\text{ } eXk)$
 $[F(V_i)]_T=\sum_j^Xf_j^i[Y(u_j)\text{ } Y(v_j)]_T;$

$\ddot{I}\text{ } f_j^i\text{ } 2\text{ } ^{\wedge}d\mu\text{`}`[Tb$
 $\text{ } \mathcal{E}iP\text{` } T\text{`QD2+iBQM } 7Q\text{`KmH}$

$$\begin{aligned}
 &(\text{ } x\text{ } x)^T([X(u)\text{ } X(v)]_T\text{ } [Y(u_j)\text{ } Y(v_j)]_T) \\
 &=(\text{ } x\text{ } x)^T(pr_1^T)(u)^T(pr_2^T)(v)^T \\
 &\quad [Y(u_j)\text{ } Y(v_j)]_T \\
 &=h(u)^T;e(v_j)^T i_{T\text{ } v;v_j} \\
 &=_{u;u_j\text{ } v;v_j}:
 \end{aligned}$$

$\ddot{I}\gg=\ddot{n}\textcircled{T}\ddot{Y}1\text{ } (RNG2KK\text{ } k\gg\mathbb{Q}\ddot{Y}\textcircled{T}\ddot{Y}1\textcircled{M}\text{ } SQBM+}\text{`}\hat{u}2[mBp\text{` } B\text{ } Mi$
 $SQBM+\text{`}\hat{u}\text{ } \mathbb{H}Biv$

$\text{„}\zeta\text{ } 8\text{ } \mathfrak{z}K\ddot{O}VrTb$
 $8\text{ } \text{†}\text{„}\zeta\text{ } j\text{ } \text{„ } 9$

$$c_{u,v}^{w;d}=\sum_i^Xa_if_{u_j}^i\text{ } v;v_j\text{ } u_j=u;v_j=v:$$

$\textcircled{R}\text{ } \acute{e}\hat{i}\mu\text{ } a_i\text{ } 2N\text{ } Q\text{` } \hat{i}\mu\text{ } f_j^i\text{ } 2\text{ } ^{\wedge}d\mu\text{ } [T\text{ } \acute{a}\grave{i}\mathfrak{a}\text{ } ^2,$
 $x_1;:::;x_r\text{ } \mathbb{Y}d\mu\mathfrak{a}\text{`}`[Tb$
 $f>\hat{i}\text{ } h?2Q\text{` } \mathfrak{g}K\mathfrak{z}\mathfrak{E}\ddot{u}b$
 $c_{u,v}^{w;d}\text{ } ^{\wedge}1\acute{e}$

NXe1Xó³ E2v AMbB;œñ£ü¥,•ó³^ 0]Ø [m MimK +Q@
 ?QKQHÖ¥wL9" Û5V[YV T`QD2+iBQM 7Q`KrBŸ¹ X X
 ¥ÜÅÜ5 b
 ¹|¹fñBÿ^V?¥\$y¹ C_{u,v}^{w,d} ¥çI #Øñß ~ 2p Hm iBQM K Tb
 ev₁;ev₂;ev₃ bYVa'F†- ñ~ F=(ev₁;ev₂) á|Û5V M wÂž X X b
 X X :` ? K TQbBxÜpâÖáì©... T@ç0[¥ËûV[•îdµ [
 T""¥LYF†b
 fñ£ü¥'ail ¿ñVü'P 0]ØĬ„Æ•¥⁻ xç~
 bWawL9" KÔ¥žŸ²T⁻–Ÿ1¿ÜÅ+...Ĭ¥e†YLBfŒ ô⁻ ŸĬ
 ¥e†Ÿ ô^"Đ-Å¥8Cb

SS1L.As

Ùs:c

Ùs:c| fÚb

RX...KĂž :G_n(C)

RXR~~UX~~X £ü a+?m#~~2~~Ti TQbBiBpB~~W~~[|BîÿĂ• G KĂž
G= :G_n(C)\$

RXK~~X~~ XBÿ/Æ¥, a†

UR3/Æ û 8² ©...ÿĂ• G ûV[3Æž ñ :G_n(C) Ĩ 3Æ^†¥O
^}"•ÿ b @™ G/P YVN3Æ3Æž :G_n/P⁰ Ĩ a+?m#~~2~~¥>
1"\$ û

$\overline{B \ B/P} \ \overline{B \ B/P} \ \emptyset \ \overline{B \ B/P^0} \ \overline{B \ B/P^0}$

Uk~~V~~T}VÍ¥f0ÿ ® (k- h?2Q`2K R~~ÿ~~X(~~B~~X~~B~~)B]T ® × a+?m#2`i
[T S (t; i) VUb'KĂž :G_n H fñ}VÍ1" ûb
Uj~~S~~QbBiB~~¥~~B~~i~~ÿ :` ? K TQbB~~B~~~~X~~B~~B~~va+?m#~~2~~TiZ7""¥d
µÿb fñÿÉ 3Æ/ ûiiÂTBî•¥f™µ TQbBiBpBivG_n L
C9µ TQbBiB~~Q~~~~p~~ÂIV :G_n f™µ M2; iBp~~B~~3Æ¥I~ V[/
Qèb

/ÆLæÿ :G_n(C) µü'¥ [T a+?m#~~2~~Ti 7BîÿĂ•Ăµf¹
z¥ [T}VÍ"db^R

kX¹...P" S₁ 7d S_n

kXR~~UX~~X Z7 T S_u S_v = ^P_w c_{u,v}^w S_w Ĩ¹...p„P" w2 S₁ 7, ^ w2 S_n \$

kXk~~i~~X X,•đy a+?m#2[~~π~~¥×çÿ ai #BHBiv

UR~~S~~V ^îµµK ë•¥i" S₁ = ^S_{n=1} S_n bĂñÂĐ w2 S₁ °µµK ñd
ÜOÊÂ 'i N P^α w2 S_N O îµ n N µw S_n Ĩb

Uk~~a~~+?m#2`iT¥3Æÿ ç m>n S_n S_m O S_w⁽ⁿ⁾(x₁;...;x_n) =
S_w^(m)(x₁;...;x_n;0;...;0)b' S_m Ĩ¥ a+?m#~~2~~Ti M x_{n+1} = x_m = 0
HKĂž S_n ¥™Tb

Uj~~V~~ È"¥×çÿ ' n @ vH YÈ n `(u)+ `(v) Z7"" c_{u,v}^w Đn í
1bfiĂ" S₁ Ĩçl¥""1-¹ îµ S_n Ĩµrb

U9~~b~~W¥1-Ôý V @™¥~ A Fl₁(C)=₁H~~B~~K_n(C) ^îµµK @™
¥Ă Kb ×ç K/ûî a+?m#2`i + ~~H~~±±aH'É~~b~~

R_U 1|1KĂž :G_n(C) \$n,c b2+i~~BQ~~M

eR

ek

XÛs:c

/Œ%« Û– u;v 2 S_n Œð S_u S_v V? #É ÑV n ¥ÂÐ w b P" S₁ E
ç |¹¥© Pª T îµ n /dBµrb
L=9ØH 89ØH áì– ñ@ v¥ S_N ïýTYÈ N = `(u)+ `(v)
y¹÷É¥ÂÐ,ö C Z7İb

jXS₁ ¥ç|Ð°4Ø³
jXRÛ5 XS₁ ç|İ¥ S i":| ^|¹i±\$
jXkKs Xi" S ¥c| f^"†,İ¥:| S_{n=1}¹ S_n VUÜîµ S₁;S₂;S₃;:::
†iîBñv"†

$$S_1 = S_1[S_2[S_3[$$

°4Ø³

S_n ° cÉ n ¥ÂÐ ' n íÂÐ
S₁ cîµµKÉ ¥ÂÐ '©iµK|ê¥ÂÐ

8è0

S₃ İ¥íí (123); (213); (132); (231); (312); (321)
S₄ İ¥íí (1234); (2134); ::; (4321)
S₁ İ¥ííV[™Â (3;1;4;2;5;7;6) iif^Bñ » eÊ-^aüÀµMÄ¥
ÂÐ L= ° # - dñÊÂ

¹|¹³¹ S₁ \$
Z7 T

$$S_u(t; \imath) S_v(t; \imath) = \sum_{w \in S_1} c_{u,v}^w(\imath) S_w(t; \imath)$$

İ ð ²TV? #1 u;v ÷v| |¥ÂÐbèÂ u 2 S₃;v 2 S₃ Œð V?Á3
w 2 S₅ b" S₁ V[dB)ØîµµKÂÐ Eç| |¥•Pb
™Tç|

$$S_1 = \prod_{n=1}^l S_n = f w : f 1; 2; ::; g ! f 1; 2; ::; g j w ^ \mu K | \hat e \neq g$$

f i Â" S₁ İ¥ÄñÂÐ° µK ñÊÂ Ðš©ÂÐ,]b

$$9X\ aF2'\mu\varnothing0\varnothing''\mu\varnothing0\neq1''$$

$$9XR\backslash X^* \ aF2'\mu\varnothing0\,,''\mu\varnothing0\mu|^{11}\acute{o}\,,uY\text{---}V[\ Bt\grave{e}0\text{---}\$$$

9XkíX X"µØ0 /BpB/2/ /Bz2`2M+2'Qā2?m@2bi + ¥ł+ýmHmb
"¿¾Bç| a+?m#½Tib aF2řµØ0V[Ø³¹ ñ"µØ0¥ ó ó Z7
""9Øİ"×1b

n5³1ü' ñ 'ÀQ
RX'ÂÐ s_i ë• S_n İ s_i i=1;::;n 1 VUŒÐÊÂ i „ i+1 ¥ '
ÂÐbèÂ

$$\begin{array}{l} S_3 \text{ İ } s_1 = (12) \text{ VUŒÐ1 } „ \text{ } 2 \text{ ' } 1237! \text{ } 213 \text{ } s_2 = (23) \text{ VUŒÐ2 } „ \text{ } 3 \\ \text{ ' } 1237! \text{ } 132b \\ S_4 \text{ İ } s_1 = (12) \text{ } s_2 = (23) \text{ } s_3 = (34) \text{ } b \end{array}$$

© i Â Ð w û V [s ^{3 1} ' Â Ð ¥ ð w = s_{i1} s_{i2} s_{ik} b
 k X ' " μ Ø 0 @ = @ ĭ i = 1 ; : : ; n 1 @ 9 : T @ ĭ l ¹

$$@ (f) = @ (f) = \frac{f(x_1; \dots; x_i; x_{i+1}; \dots; x_n)}{x_i \quad x_{i+1}} \quad f(x_1; \dots; x_{i+1}; x_i; \dots; x_n)$$

T " ^ 5 Ü f " İ ¥ x_i „ x_{i+1} Ą Ð Ð ð f " M h K ^a " [x_i x_{i+1} b
 è 0 ! f = x₁ 5 @ (x₁) = $\frac{x_1 - x_2}{x_1 \quad x_2} = 1$ b
 n = 3 ¥ è 0 S₃ İ ¥ Â Ð V [" B » " F V U Â 123 V U š © Â Ð 321 V U K É
 Â Ð w₀ b

$$U R @ (x_1) = 1 \quad y^1 \quad @ \quad \text{Ð} x_1 \text{ „ } x_2 \text{ }^a M h \square x_1 \quad x_2 \text{ " } [\quad x_1 \quad x_2 \square 1 b$$

$$U k @ (x_2) = 1 \quad y^1 \quad @ (x_2) = \frac{x_2 \quad x_1}{x_1 \quad x_2} = 1 b$$

$$U j @ (x_1^2) = \frac{x_1^2 \quad x_2^2}{x_1 \quad x_2} = x_1 + x_2 b$$

$$U 9 @ (x_1 x_2) = \frac{x_1 x_2 \quad x_2 x_1}{x_1 \quad x_2} = 0 b$$

$$U 8 @ (x_2) = 1 \quad \ddot{E} \gg^1 \quad @ \quad \text{Ð} x_2 \text{ „ } x_3 b$$

$$U e @ @ (x_1) 5 T " \quad @ \quad \text{R} \quad \text{ĭ} \quad x_1 \text{ , } G \quad \text{ĭ} \quad x_2 ; x_3 \quad @ (x_1) = x_1 \quad T " \quad @ \quad @ (x_1) = 1 b$$

$$U d @ @ (x_1 x_3) = @ (x_1 x_3) = x_1 + x_2 b$$

n = 4 ¥ è 0 S₄ İ μ 4 ! = 24 ñ Â Ð b ! w = 3412 V U † (1; 2; 3; 4) 7 ! (3; 4; 1; 2)
 V s ^{3 1} w = s₁ s₂ s₁ s₃ V . ^a P b

$$U R @ (x_1^2 x_3) = (x_1 + x_2) x_3 b$$

$$U k @ @ (x_3) = @ (x_3) = x_3 \quad y^1 \quad @ \text{ , } \text{ĭ} M \text{ , } c \quad x_1 ; x_2 \text{ ¥ } [b$$

$$U j V \text{ĭ} \quad w = s_1 s_2 \quad \mu \quad @_{s_1 s_2} = @ @ b 9 \emptyset @ @ (x_1)$$

$$@ (x_1) = x_1; \quad @ (x_1) = 1$$

$$\hat{I} [\quad @_{s_1 s_2} (x_1) = 1 b$$

$$U 9 V \text{Ĭ} \quad \hat{A} \hat{D} \quad w = 4321 = s_1 s_2 s_3 s_1 s_2 s_1 \quad K \acute{E} \hat{A} \hat{D} \quad w_0 \quad \mu$$

$$@_0 = @ @ @ @ @ @$$

$$T " \quad \dagger [T \quad \ddot{o} \square \check{z} \quad \ddot{e} \quad [T b$$

Â Ð < ¥ " μ Ø 0 @ ĭ © i Â Ð w 2 S_n ĭ l @ ¹ a ' F † ¥ ' " μ Ø 0 b
 è Â w = s_s H @ = @ @ b

$$a F 2 r " \mu \emptyset 0 \quad @_{w/v} \quad \text{ĭ} l^1 \quad @_{w/v} = @ \quad @^1 \quad \text{İ} \quad @^1 \wedge @ \text{ ¥ } l \emptyset 0 \quad V U \emptyset 0$$

$$- \dagger b f \text{ , } ^\wedge e \dagger \text{ ¥ } f " " E \quad 7^{\wedge} 5 \quad f " " \quad @ \text{ ¥ } l \quad \emptyset \quad " \quad @ \quad T " b$$

$$^1 l^1 f " \text{ĭ} l \$ \quad f \text{ñ} \text{ĭ} l \div \text{ĭ} \quad a + ? m \# 2 ` i + \text{İ} \text{Z} \text{7} \text{m} \text{İ} \text{¥} \text{M} \text{b} \text{b} \text{ ' } \acute{a} \text{İ} X |$$

$$\text{ñ} [T Z 7^1 \quad a + ? m \# \text{1} \text{Ti} \text{¥} L \ddot{Y} F \dagger H \quad a F 2 \text{ř} \mu \emptyset 0 4 \quad B \ddot{O} " d \text{¥} 9 \emptyset Z$$

$$E b \quad 8 \ddot{Y}^a \quad \hat{A} T \quad S_w \quad ^\wedge \quad a + ? m \# \text{1} \text{Ti} \text{ } ^* 1 \quad @_{w/v} (S_u) \quad \acute{o} \quad F \dagger " " \quad c_{u,v}^w b$$

$$n = 3 \text{ İ } \text{¥} \quad a F 2 r " \mu \emptyset 0 \text{ è } 0$$

$$U R @_{s_1/s_1} = @ \quad y^1 \quad @_{s_1} \quad @_{s_1}^1 = @ \quad \text{š} \text{ © } \emptyset 0 b$$

$$U k @_{s_1 s_2 / s_2} = @_{s_1} \quad @_{s_2}^1 \quad 5 T " \quad @_{s_2}^1 \quad @_2 \text{ ¥ } l \quad T " \quad @_1 b$$

$$U j \emptyset \emptyset @_{s_1 s_2 / s_2} (x_1)$$

$$@_2 (x_1) = x_1 \quad y^1 \quad x_1 \text{ , } G \quad x_2 ; x_3$$

$$\hat{I} [\quad @_{s_2}^1 (x_1) = x_1$$

$$T " \quad @_1 \quad @_1 (x_1) = 1$$

$$y N @_{s_1 s_2 / s_2} (x_1) = 1 b$$

$$U 9 V \text{Ĭ} \quad w = w_0 = 321 \quad v = s_1 = (12) \quad \mu \quad @_{w_0/s_1} = @_{w_0} \quad @_{s_1}^1 b$$

$$\text{ , } \acute{o} \text{ İ} \quad a F 2 \text{ř} \mu \emptyset 0 " \text{ĭ} 9 \emptyset Z 7 " " \quad c_{u,v}^w \quad \check{z} \ddot{Y}^{\wedge} \quad E B ` B \text{İ} \text{X} \text{¥} \text{p} \cdot \ddot{U} 5 b$$

8 X I " μ Ø 0 ¥ ç I

8 X R U 5 X I " μ Ø 0 ^ Â ... ç I ¥ \$

8 X k í X X I " μ Ø 0 @ ¹ ^ @ ¥ I Ø 0 ' i @ @ @ ¹ = @ ¹ @ = B / š © Ø
0 b

Â ... ç I @ ¹ \$ ® ¿ @ , ^ I Ø 0 ³ 1 Y V ð Z T ç I b L = 9 Ø ï
á ì Y È " [/ Z T Ø ³

Z T B T ¹ Ø 0 ¥ ™ T I | @ ¹ j ¹ B ñ ™ T Ø 0 i @ @ @ ¹ = 1 b f F † "
Đ ï È " ¿ 3 î f " ¥ 9 Ø b

Z T = Y V [T ì [T ì Z [x ₁ ; :: : ; x _n] ï @ ¥ ^ c î μ ë [T y
N @ ¹ V [Ø ^{3 1} ì ï ¥ I b

n = 3 ï ¥ è 0

U R Q (x ₁) = 1 î [@ ¹ (1) = x ₁

ee

XÛs:c

NXËbW 1h¥°\$Ø³

NXRUX R¹|¹^Ú»oë¥ K\$ X1hçlĭ¥ óÚ»oë K ó¹¹cl\$
¹|¹1" K\$,?°α" Bñoë—\$

NXkX RXlh³1]Hj@ ñ°M± ¥Hq

URIM®T" • T T" ë,?μ©...,îÄ

UkVêŸ %ê 1^BñÄ Àμ óó ó

¹|¹,?"μK»oë\$μK»oë S^k μó ,^Vê¥bèÂ S³ μBñ j »¥
óó ób

K¥cl

S¹ S³ S⁵ S^{2m+1} ! S¹

S³ ĩ μ j »ó S⁵ ĩ j »ó\$A ÆÁ3 8»ó S^{2m+1} ĩ îμ®
¿ 2m+1 »¥óû\$AÜ ' m!1 H ó\$w_íkÚ»b ¿©... μK»4©€
]ØØ, S¹ A Ÿü^Àμ©...óiiñ^Vê¥

NXjUX5 k¹|¹ X ET ü^1®¥\$ Xfª¥°\$^¹¹\$

NXgX kX1®T"¥çl ÂT t (x;e)=(x;e) *¹ t A¶^†Êíb

1o 1h ¥T"¹¹®¥ii" †Êí Àμ©... t ?ž t e=eb

'P x ^X ĩ¥,îÄ à@ P "æü÷%díŸĭx—pPñe"l—0

RkX >BH#2ðÌÙ5Đ a+?m#2Ø

RkX~~Ø~~¥åN ß X >BH#2E ò Ù5 ^ >BH#2 RNYjy Ø Ó S="Đ Ev
 ö 4 ¥ kjñ Ù5İ¥» Rñ ð Ó V•¹
 óÜ aQT?m~~¥~~6Xi2'ð Æ MĐ•¥À Qi i<"¿ s İ¥ Ù5 +Y^{^1}¿
 "†Bf"çl,M ¥ Ù5iiyë ø á¥"Đ \$- b ò
 +... Ô â/ >BH#2E ð Ù51p¹ a+?m#2Ø a+?m#2`i + y~~He~~amHmb
 ì¥"Đ \$b a+?m#2Ø†®£S"Đ E >2`K MM a+?m#2ØV@ ?Z¥B*
 9Ø}" +... Æ Äñ"¥Z E b

RkXkX a+?m#2`i Ù5 Xa+?m#2Ø i÷¿ [/9" Ù5 ó ç Ü ë İ r
 H°L,, s ñ Ä °LĐ Ä ¥ Æ Äñ" ^ \$'Hq ó B î Ê Â ó H s Ä ^'ç ¥ b Æ
 a+?m#2Nw<ž ÷ Ú »
 ! F_{I_n}(C) ¹ @™ w 2 H^{2(w)}(F_{I_n}(C)) ¹ a+?m#2`i w 2 S<sub>n</sub> `(w) ¹ Â Đ
 É b @™]Ø İ ¥² ¹

U9dV H (F_{I_n}(C); Z) = Z[₁; ₂; ::; _{n-1}]/ U`2H iBQMbV

a+?m#2Ø¥,• Ù5^ ó ç Ø ñ a+?m#2`i; v; w ñ İ ¥ ð Z7""
 ' a+?m#2`È" c^w_{u,v} Ä ... 9Ø\$

U93V
$$u \quad v = \frac{X}{w} c_{u,v}^w$$

RkXjX a+?m#2`i ¥ F† i l X+... c<sup>w</sup><sub>u,v</sub> 9"¥ ^ @™ İ Ø ñ a+?m#2`i
 8¥ Æ Äñ"

U9NV
$$c_{u,v}^w = OX(u) \setminus X(v) \setminus X(w)$$

İ X(w) VU @™ İ ® Â Đ w % ç ¥ a+?m#2`i
 +Y¹ c^w_{u,v} 2 N ^ d μ œ " i i f ü ^ a+?m#2`i"¥ ž Ÿ T Q b B i B p B i v

RkX~~Ø~~À è 0 :` b b K M M B K#2H B K e t ¥ f f ^ :` b b K M M B M
 Gr(k;n) k » 0 b W î ¥ @™ b N H a+?m#2`i ® T ` i B i B Q (M b ₂
_k 0) ì „ b : B K#2H B

U8yV
$$= /2(\quad _i \quad _{i+j})_{1 \quad ij \quad k}$$

İ _{i i+j} ^ + y ¥ a+?m#2`i „¹ μ H |¹, b

RkX8X >BH#2 Ù5¥³% Xa+?m#2`i S9"Z E ¿ ó + y Ê Â ð Ø ó
 T`BM+BTH2 Q7 b Æ 2+Ø B £ H B °ž i B Q M N M y M / 2` b 2 M ; 2 M, b Ø M M /
 Æ £ ü a+?m#2Ø¥9"?5^>!¥ i i' @™ ¥ B î Ê Â H q / a+?m#2`i
 8¥ Æ Äñ" © ¿ ® F† T9Ø¥²T b
 f ž T Æ á >BH#2E ð Ù5¥³% b

RkX~~e~~X Ù5 S Q b B i B B i v XĐ59">!Ÿ X³% Æ a+?m#2`i
 È"¥ ž Ÿ c^w_{u,v} 0 Ù5 „ ÷ ' a ¥ " Đ

U R W... ž Ÿ c^w_{u,v} 9" Æ Äñ" +... 1 - ^ d μ ¥
 U k W ž Ÿ a+?m#2`i T i ¥ " ^ ž 9 ^ d μ ¥ \$
 U j W † ž Ÿ ^ ž i F† ~ ° ¢ ó ž Ÿ £ ü \$

R 9 X0] Ø ¥ ç l

e N

:` ? K S Q b B i B p B i v ž h 1 2 Q B 2 K 5 ¥ × 1 Ú ñ b ñ £ ü :` b b K M @
M B f M / Z 7 " " V [• î ž " " ¥ † [T F † b 7 : Q s B Q ¥ ; T ' Ó ¥ ö
5 | f B ² T w < ž Ø × M " ¥ f T M b

R j X S₁ ¥ è 0

R j X R U X X S₁ V [ñ è 0 — \$

R j X K X X S₁ ^, Ó Ĭ B ñ “ \$ Æ × 1 ¥ À Q b, Ó ð Ó ¥ ç l

S₁ / 2 M Q i 2 b i ? 2 m M B Q M Q 7 H H } M S₁ i 2 T_{n=1}^{S₁₂} S_n m r ? B Q M b - B X 2 X -
+ M # 2 m M / 2 ` b i Q Q / b i ? 2 b v K K 2 i ` B + ; ` Q m ; 2 ; Q : m X i ? 2 + Q m M i # F

° 4 Ø ³

S_n ^ n ñ í í f 1 ; 2 ; : : ; n g ¥ ë • b è Â

S₁ = f 1 g š © Â Ð

S₂ = f 1 2 , 2 1 g k ñ Â Ð

S₃ = f 1 2 3 1 3 2 2 1 3 2 3 1 ; 3 1 2 3 2 1 g e ñ Â Ð

S₄ μ 4 ! = 2 4 ñ Â Ð

S₁ ü ^ Ü î μ S₁ ; S₂ ; S₃ ; : : : † î î B ñ í K " †

S₁ = S₁ [S₂ [S₃ [= f î μ μ K ¥ Â Ð g

¹ | ^{1 3} 1 S₁ \$
Z 7 T

$$S_u(x;y) \ S_v(x;t) = \sum_{w \geq S_1}^X c_{u,v}^w(y;t) \ S_w(x;t)$$

İ ð ² T V ? # 1 u ; v ÷ v | | ¥ Â Ð b Y V S₁ á ì V [d B) Ø î μ μ K Â
Ð , „ • ó w < ç ' ñ S_n Ó ¥ Ü 5 b
è

w = (3 ; 1 ; 4 ; 2 ; 5 ; 5 ; : :) 2 S₁ V U - 9 Ê Æ Ð Â Ê Â , M b
ÿ i S₁ , ^ í K “ • 7 ^ B ñ ó Å μ K Ó ¥ i " i i ñ ° c μ K | ê ¥ Â Ð '
° μ μ K ñ Ê Â , ^ š © ¥ b f Ð S₁ T ¹ V " í K " ¥ ë • ^ ,] ¥ À Q b

R 9 X0] Ø ¥ ç l

R 9 X B X X ° 0] Ø ¥ ç l b

R 9 X k X X 0] Ø U Z m M i m K * Q ? Q K Q H Q ; j Ø \$ „ Æ w L
9 " ¥ × 1 w < b
, • ± X ô Y] Ø H (X) 9 " ¥ ^ X 0 [¥ M Æ ñ " b 7 0] Ø Q H (X)
É B „ l n ´ E μ Ø w L ¥ ñ " b
ç l ! X ^ ; Á ¯ @ T M 2 H₂ (X ; Z) V U w L Ë b 0] Ø ¥ ð E ç l ¹

[] ? [] = q H [] ; [] ; [] i_{0;3} ; []

İ

q ^ T M T M : c w L Ë ¥ "

h i_{0;3} , ^ : ` Q K Q p @ q B i M M 9 " i @ + ç H q ¥ • " Ä w L ñ "

dy

XÛs:c

Đ ô Y] Ø ¥ u Y P
ô Y] Ø u v = P_w c^w_{u,v} w
0] Ø u ? v = w; q c^w_{u,v} () w
u Y ¿ " " c^w_{u,v} () , Ç G ¿ 0 [Î G ¿ ð ñ ì ¥ w L Ë b
+ ... ÷ : ` Q K Q p @ q B M i 2 M p Ø Ì ¥ % ê Æ , U h 6 h v ÿ \$ E Q M i b 2 p B + ?
© ! „ Æ } " + ... b ñ 9 " ¥ ^ { S : Ä ¥ × ç ~ ¥ ñ " ^ ô Y M Æ Ø , ¥ ó w L ñ
' ó b

R 8 X h ? 2 Q ` 2 K R X R U h ` B T H 2 a + ? m # 2 ` i S Q b B i B p B i v
h ? 2 Q ` 2 K R X R U h ` B T H 2 a + ? m # 2 ` i ; S Q b B i B p B i v i 2 / / 2 b + 2 M i b
q 5

$$S_u(x;y) \ S_v(x;t) = \sum_w c_{u,v}^w(y;t) \ S_w(x;t)$$

İ c^w_{u,v}(y;t) 2 N[t_i y_j] ^ y „ t ¥ ž " " [T b

R 8 X B X S₃ İ £ b 2 T ` i 2 / / 2 b + 2 M i b K t . 2 (b) K B M (v b è Â | n = 3 M ¹ x₁; x₂; x₃
„ F - “ M y₁; y₂; y₃ Đ t₁; t₂; t₃ b
a 2 T ` i 2 / / 2 b + 2 M i b K t . 2 (b) K B M (v b è Â |

$$u = 132 \quad (.2(u) = f2g); \quad v = 321 \quad (.2(v) = f1; 2g)$$

5 K t . 2 (b) = 2 K B M (v b 1 , î ë < ¾ ^ 2 1 , i @ b
ž ' ¥ è 0 |

$$u = 213 \quad (.2(u) = f1g); \quad v = 231 \quad (.2(v) = f2g)$$

5 K t . 2 (b) = 1 K B M (v b 2 i @ H q

R 8 X k 8 9 Ø £ X S₃ İ X © a + ? m # 2 T i ® " µ Ø 0 ¾ w ç l

$$S_{123} = 1; \quad S_{213} = x_1; \quad S_{132} = x_2$$

$$S_{231} = x_1 x_2; \quad S_{312} = x_1 + x_2; \quad S_{321} = x_1^2 x_2$$

× a + ? m # 2 T i ¥ Ÿ É

$$S_w(x;0) = S_w(x)$$

$$S_{w_0}(x;y) = \prod_{i=1}^n (x_i - y_i) \quad \text{İ} \quad w_0 = (n; n-1; \dots; 1)$$

$$\zeta \ w = 321 \ K \acute{E} \hat{A} \grave{D} \quad S_{321}(x;y) = (x_1 - y_1)(x_2 - y_2)(x_3 - y_3) \ b$$

| u = 213 \ v = 231 \ i @ \ b 2 T ` i 2 / / 2 b } Æ M i b T

$$S_{213}(x;y) \ S_{231}(x;t) = \sum_w c_{213,231}^w(y;t) \ S_w(x;t)$$

9 Ø P H

$$S_{213}(x;y) = x_1 - y_2; \quad S_{231}(x;t) = x_1 x_2 - x_1 t_3 - x_2 t_3 + t_2 t_3$$

ð ¹

$$(x_1 - y_2)(x_1 x_2 - x_1 t_3 - x_2 t_3 + t_2 t_3)$$

Z 7 ^a ò [" " û ^ y „ t ¥ ž " " [T Â y₂ t₃ © £ ž Ÿ b

RdX ELIhaPL@h P ŹkyyM ...il

d R

R8XpXBñè0 y=t=0 |Äf™ X' y=t=0 H h?2Q`2K|ÄXÜÄ¥
: ` ? K SQbBiBpBiv

$$c_{u;v}^w(0;0)=c_{u;v}^w\,2\,N$$

NHü^ôY¥ a+?m#²2Èi" 9" @™İ a+?m#B¥ÆÄñ"b
èÂ Fl₃(C) İ

$$S_{213}\,S_{231}=S_{321}$$

$$y^1\,_{213}\,_{231}=_{321}$$

RNXW% s X RkX@: ` b bK MMB¥M... il
 Bñ k@: ` b bK MMBwM< : ` b bK MMB™M(k;n) ĩ¥Bñ a+?m#2`i
 8b^kçl1p

$w(1) < w(2) < \dots < w(k) \leq w(k+1) > w(k+2) > \dots > w(n)$
 + ... f i Å" w ¥ / t Ä" t .2(w) z ^ fkgii°µ Bñ /2b+2MBi
 Ê Â k b f ñ t B /2b+2MBi s î ñ ó¾49v ó
 - k Ê ø ï¾49) < : ` b bK MMBBMM ~ ;Hq V₁ V₂ V_k
 a n k Ê ø ï¾4h) < : ` b bK MMBBMM c 1"
 è Â w=(1;3;4;2)2 S₄ ^ k@: ` b bK MMB M
 - k Ê 1<3¾49
 a k Ê 4>2¾4h
 < : `(2;4) ĩ¥ a+?m#B`i

kX₁;k₂ ¥T"
 k₁ „ k₂ S: ñ : ` b bK MMBBMM á i ñ ñ %ç <¥ a+?m#B`S Y
 Ê ç ' Ö : ` b bK MMBBMM b
 ' u ^ k₁@: ` b bK MMBBMM k<sub>2</sub>@: ` b bK MMBBMM <

$S_u(t; v) = s^{(1)}(t/v); S_v(t; i) = s^{(2)}(t/i)$
 ĩ s ^ a+?m" (1); (2) ^ Ð k₁;k₂ M1¥s·b^j
 jX¹I¹žŸG ç : ` b bK MMBBMM i i y 0 Ä Å
 ' u;v û ^ : ` b bK MMBBMM ñ M a+?m#2TIV[y0Ä<sup>1</sup> 7 +iQ`B+ñm`
 f"

$S_u(t; v) = s(t/v)$
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 $\text{ } \text{ } \acute{D} \acute{y} - \text{ } u \neq \hat{1} \mu / \text{ } \text{ } \check{A} \hat{u} \text{ " } \check{I} \text{ } p \neq P \S \text{ } v \neq \hat{1} \mu / \text{ } \text{ } \check{A} \hat{u} \text{ " } \check{I} \text{ } p \neq . \S b$

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25. $B^-(w)$ 与 B^- 的关系

25.1. 问题. $B^-(w)$ 是什么? 它和 B^- 有什么关系?

25.2. 回答. 定义回顾:

根据定义 1.23 (子群 $N^-(w)$ 与 $B^-(w)$), 我们有:

$$N^-(w) := N^- \cap wN^-w^{-1}$$

$$B^-(w) := T \cdot N^-(w)$$

各符号的含义:

- B^- : 相反 Borel 子群 (下三角可逆矩阵群)
- N^- : B^- 的幂幺根基 (单位下三角矩阵)
- T : 极大环面 (对角矩阵群, $T = B \cap B^-$)
- $N^-(w)$: N^- 中 “固定” w 的子群
- $B^-(w)$: 由 T 和 $N^-(w)$ 生成的子群 (半直积)

极端情形:

- 当 $w = w_0$ (最长元) 时: $B^-(w_0) = T$ (因为 $N^-(w_0) = \{e\}$)
- 当 $w = e$ (单位元) 时: $B^-(e) = B^-$

几何意义:

$B^-(w)$ -不变的闭链在 Schubert 胞腔的相交理论中扮演重要角色。直观上:

- (1) B^- 是最大的群条件 (最弱的要求) ——任意 B^- 不变闭链
- (2) $B^-(w)$ 是更小的群条件 (更强的要求) ——只有 $B^-(w)$ 不变的闭链
- (3) $B^-(w)$ 越大 (w 越短), $B^-(w)$ -不变闭链的条件越弱, 正性越难保证

为什么 Refined Graham Positivity 更强:

原始 Graham 要求 B^- -不变闭链, 而 Refined 版本要求 $B^-(w)$ -不变闭链。条件更弱 (更容易满足), 但结论反而更强——根被限制在 $I(w)$ 而非所有正根。这似乎矛盾, 但实际上是因为 Refined 版本通过 separated descents 条件额外保证了横截相交性。

26. $\mathbb{N}[-\alpha]_{\alpha \in I(w)}$ 的含义

26.1. 问题. 在 Refined Graham Positivity 的证明中, 系数 $c_{u,v}^w(\mathbf{y}, \mathbf{t}) \in \mathbb{N}[-\alpha]_{\alpha \in I(\tau)}$ 这个符号是什么意思? 为什么用 $-\alpha$ 而不是 α ?

26.2. 回答. 符号分解:

记号 $\mathbb{N}[-\alpha]_{\alpha \in I(\tau)}$ 表示 “以 $\{-\alpha \mid \alpha \in I(\tau)\}$ 为变量的非负整数系数多项式”。

更仔细地:

- $\mathbb{N}[x_1, x_2, \dots, x_k]$: k 个变量的非负整数系数多项式环
- $\{-\alpha \mid \alpha \in I(\tau)\}$: 变量集合

具体例子:

由第 3.2 行的计算, $I(\tau) = \{y_j - t_i \mid 1 \leq i, j \leq n\}$, 因此

$$-\alpha = -(y_j - t_i) = t_i - y_j$$

所以 $\mathbb{N}[-\alpha]_{\alpha \in I(\tau)} = \mathbb{N}[t_i - y_j]_{i,j \geq 1}$, 这正是 Graham Positivity 断言的系数形式。

为什么是 $-\alpha$?:

- (1) **符号约定:** 在等变上同调中, T 的权通常用负根表示。具体来说, $H_T^*(\mathbf{pt}) = \mathbb{Z}[t_1, \dots, t_n]$ 中 t_i 对应负根 $-\epsilon_i$ 。

(2) **Corollary 2.4 的形式**: 第 1.2 行的 Corollary 2.4 给出

$$[Y]_T = \sum_{w \in W} N[-\alpha]_{\alpha \in I(w)} \cdot [\overline{B^- w B / B}]_T$$

这里 $N[-\alpha]$ 表示系数是关于负根的非负多项式。

(3) **链式法则**: 在证明中 (第 3.2 行), 因为交集在 $N^-(\tau)$ 下闭, 而 $N^-(\tau)$ 的李代数与 $I(\tau) = \{y_j - t_i\}$ 相关, 所以最终系数出现在 $N[-\alpha]_{\alpha \in I(\tau)}$ 中。

直观理解:

这是把”关于 $t_i - y_j$ 的非负整数系数多项式”写得更加抽象和紧凑的形式。两种写法完全等价:

$$N[-\alpha]_{\alpha \in I(\tau)} = N[t_i - y_j]_{i,j \geq 1}$$

27. EQLR 系数的定义与意义

27.1. 问题. EQLR 系数是什么? 它在等变量量子量子上同调中扮演什么角色?

27.2. 回答. **EQLR = Equivariant Quantum Littlewood-Richardson (等变量量子 Littlewood-Richardson)**

EQLR 系数 $c_{u,v}^{w,d}$ 是等变量量子量子上同调 $QH_T^*(X)$ 中的结构常数, 由以下展开式定义:

$$\sigma(u)^T \circ \sigma(v)^T = \sum_{w,d} q^d \cdot c_{u,v}^{w,d} \cdot \sigma(w)^T$$

其中:

- $\sigma(u)^T$ 是 T -等变 Schubert 类
- q^d 是量子参数 (对应 Gromov-Witten 不变量)
- d 是多重次数 (quantum degree)

几何含义:

从几何上看, EQLR 系数由等变 Gromov-Witten 不变量定义为:

$$c_{u,v}^{w,d} = \pi_{\star}^T ((ev_1^T)^*(\sigma(u)^T) \cdot (ev_2^T)^*(\sigma(v)^T) \cdot (ev_3^T)^*(\tilde{\sigma}(w)^T))$$

其中 $\overline{\mathcal{M}}_{0,3}(X, d)$ 是稳定映射的模空间, ev_i 是评估映射。

Mihalcea 定理 (Theorem 1):

Mihalcea 证明了: 设 $u, v, w \in W^P$ 且 d 为多重次数, 则 EQLR 系数 $c_{u,v}^{w,d}$ 是关于负单根系 $x_1, \dots, x_r$ 的非负系数多项式。

与经典 LR 系数的区别:

- 经典 LR 系数 $c_{u,v}^w$: 非负整数 (计数的几何相交点数)
- 等变 LR 系数 $c_{u,v}^w(y, t)$: 关于 $t_j - y_i$ 的非负多项式
- EQLR 系数 $c_{u,v}^{w,d}$: 关于负根 $x_1, \dots, x_r$ 的非负多项式 (带量子参数 q^d)

证明思路:

Mihalcea 的核心思想是**归约**——将 EQLR 系数的计算归约到已经在 Graham Positivity 定理中解决的问题。具体步骤:

- (1) 考虑空间 $X \times X$ 及其半直积 $B' = T \cdot (U^- \times U^-)$ 的作用
- (2) 利用投影公式将 EQLR 系数的计算归约到 $X \times X$ 上的计算
- (3) 应用 Graham 的正性定理 (Proposition 6.1): $[V]_T$ 可以唯一表示为 $[Y(u) \times Y(v)]_T$ 的非负系数线性组合
- (4) 使用等变 Poincaré 对偶完成证明

为什么重要：

EQLR 系数的正性性质是等变量量子量子同调的核心结构。它连接了：

- 经典 Schubert calculus (cohomology)
- 等变上同调 (equivariant cohomology)
- 量子同调 (quantum cohomology)

这三种理论在旗流形的 Schubert calculus 中统一起来。⁹

28. 双重 Schubert 多项式的除差算子定义

28.1. 问题. 双重 Schubert 多项式 $\mathfrak{S}_w(\mathbf{x}; \mathbf{y})$ 如何用除差算子定义？

28.2. 回答. 定义 (参考 Kirillov-Maeno [11]):

$$\mathfrak{S}_w(\mathbf{x}, \mathbf{y}) = \partial_{w^{-1}w_0}^{(x)} \prod_{i+j \leq n} (x_i + y_j)$$

其中：

- $\partial_{w^{-1}w_0}^{(x)}$ 表示只在 x 变量上施加除差算子
- 顶部多项式 (top polynomial) $\prod_{i+j \leq n} (x_i + y_j)$ 对应最长置换 w_0

与单重情形的对比：

情形	定义公式	顶部多项式
单重	$\mathfrak{S}_w(x) = \partial_{w^{-1}w_0}(x^\delta)$	$x^\delta = \prod_{i=1}^{n-1} x_i^{n-i}$
双重	$\mathfrak{S}_w(x, y) = \partial_{w^{-1}w_0}^{(x)} \prod_{i+j \leq n} (x_i + y_j)$	$\prod_{i+j \leq n} (x_i + y_j)$

关键性质：

- (1) $\partial_i^{(x)}$ 只作用在 x 变量上, y 变量保持不变
- (2) 递推关系: $\mathfrak{S}_{ws_i}(x, y) = \partial_i^{(x)} \mathfrak{S}_w(x, y)$ (当 i 是 w 的 descent 时)
- (3) 对称性: $\mathfrak{S}_w(x, y) = (-1)^{\ell(w)} \mathfrak{S}_{w_0 w}(y, x)$
- (4) 退化: 当 $\mathbf{y} = \mathbf{0}$ 时, $\mathfrak{S}_w(\mathbf{x}, \mathbf{0}) = \mathfrak{S}_w(\mathbf{x})$

S_2 的具体例子：

在 S_2 中, $n = 2$, $w_0 = 21$ 。

- $\mathfrak{S}_{21}(\mathbf{x}, \mathbf{y}) = (x_1 + y_1)(x_1 + y_2)(x_2 + y_1)$
- $\mathfrak{S}_{12}(\mathbf{x}, \mathbf{y}) = x_2 + y_1$

除差算子的几何意义： 除差算子 ∂_i 在几何上对应于沿第 i 个超平面进行截面, 递归地将高维相交问题降维。¹⁰

29. $N^-(w)$ 与 Schubert 胞腔 B^-wB/B 的同构

问题. 为什么 $N^-(w)$ 同构于 Schubert 胞腔 $B^-wB/B \cong \mathbb{C}^{\ell(w_0) - \ell(w)}$? 请解释几何直观、代数同构映射的意义, 以及为什么这个同构说明 $N^-(w)$ 是连通的。

⁹问: EQLR 系数是什么? 见附录 section 27。

¹⁰除差算子的几何意义见附录 section 24。

回答. 1. 几何直观: 维数公式 $\dim N^-(w) = \ell(w_0) - \ell(w)$

N^- 是单位下三角矩阵群, 维数为 $\binom{n}{2}$ (下三角矩阵的非对角元有 $\binom{n}{2}$ 个自由参数)。作为簇, N^- 同构于 $\mathbb{C}^{\binom{n}{2}}$, 因为每个元素可以唯一写为根子群的乘积:

$$N^- \cong \prod_{\alpha \in \Phi^-} U_\alpha$$

其中 $U_\alpha \cong \mathbb{C}$ 是对应负根 α 的单参子群。

$N^-(w) = N^- \cap wN^-w^{-1}$ 的维数取决于哪些负根 α 满足 $w(\alpha) > 0$ (即 w 把负根变成正根)。关键在于:

$$N^-(w) = \prod_{\substack{\alpha \in \Phi^- \\ w(\alpha) > 0}} U_\alpha$$

负根是 $\Phi^- = \{-\epsilon_i + \epsilon_j : 1 \leq j < i \leq n\}$, 共 $\binom{n}{2}$ 个。 $w(\alpha) > 0$ 的负根 α 个数等于 $|J(w)| = \binom{n}{2} - \ell(w)$, 其中 $J(w) = \{(i, j) : i < j, w(i) < w(j)\}$ 是 w 的非逆序集。

因此:

$$\dim N^-(w) = \binom{n}{2} - \ell(w) = \ell(w_0) - \ell(w)$$

几何图像: N^- 本身同构于最大的相反 Schubert 胞腔 $B^-w_0B/B \cong \mathbb{C}^{\binom{n}{2}}$ 。而 $N^-(w)$ 是 N^- 的一个“切片”, 只保留那些 w 不翻转的负根方向。

2. 代数同构: $N^-(w) \cong B^-wB/B$

构造同构映射:

$$\phi : N^-(w) \rightarrow B^-wB/B, \quad x \mapsto xwB/B$$

验证是良好定义的: 若 $x \in N^-(w)$, 则 $x \in N^- \subset B^-$, 所以 $xwB \in B^-wB$, 陪集 xwB/B 落在 B^-wB/B 内。

单射性: 若 $xwB/B = ywB/B$, 则 $w^{-1}x^{-1}yw \in B$ 。同时 $w^{-1}x^{-1}yw \in w^{-1}N^-w \cap N^- = N^-(w)$ 。但 $B \cap N^-(w) = \{e\}$, 所以 $x = y$ 。

满射性: 任取 B^-wB/B 中元素。由于 $B^- = N^-T$, 任意元素可写为 $n_1twB = n_1w(w^{-1}tw)B$ 。但 $w^{-1}tw \in T \subset B$, 所以实际上 $n_1twB = n_1wB$, 且 $n_1 \in N^-, n_1 \in N^-(w)$ 。

3. 同构映射 $x \mapsto xwB/B$ 的意义

G/B 是齐性空间, B^-wB/B 是过基点 wB/B 的胞腔。映射 $x \mapsto xwB/B$ 把 $N^-(w)$ 中的元素 x 看作“旗的位移”——把基旗 wB/B 推到旗 xwB/B 。

$N^-(w) \cong \mathbb{C}^{\ell(w_0) - \ell(w)}$ 提供了胞腔 B^-wB/B 的自然坐标系。每个 $x \in N^-(w)$ 指定了胞腔中一个唯一点。

从齐性空间角度看, wB/B 的稳定化子是 B , 而 $N^-(w)$ 是那些“与 w 的相对位置在 N^- 中”的元素构成的子群。

4. 连通性

N^- 是幂么群 (unipotent group), 指数映射 $\exp : \mathfrak{n}^- \rightarrow N^-$ 是双射 (代数群情形的对数映射)。因此 N^- 微分同胚于其李代数 $\mathfrak{n}^- \cong \mathbb{C}^{\binom{n}{2}}$, 是连通的。

$N^-(w)$ 是 N^- 的闭子群, 因此作为代数群也是连通的。

同构 $N^-(w) \cong \mathbb{C}^{\ell(w_0) - \ell(w)}$ 说明 $N^-(w)$ 是仿射空间。仿射空间是连通的 (甚至单连通)。

参考文献: Humphreys, [9, Section 28]。¹¹

¹¹问: 为什么 $N^-(w) \cong B^-wB/B$? 见附录 section 29。

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